JointZip: Implementation Details and Complete Ablation Results

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ABSTRACT

This companion records all aggregate results of the completed six-backend E/ER/EA/ERA and fixed-candidate-pool correction ablation. It accompanies the research manuscript's interpretation and preserves all prescribed TEST settings, task families, and unfavorable results. It is experimental documentation, not an additional page allowance for the research-paper submission. Historical grid-version results are not pooled here.

A1 CORRECTION REPRESENTATION DETAILS

These details specify the unchanged correction formats used by the evaluated framework. They complement the main paper's explanation of replacement selection and replay.

A correction count is not a storage cost. For each repaired vector, we compare the applicable representations in Table A1, including all masks, scales, exceptions, tags, and lossless-compression overhead. Quantized formats use target scales when available; format 0 requires reproducible double-residual addition.

Position and exception masks use one bit per channel-major/time position; format 5 uses one bit per channel. Bits are least-significant first within each byte, and selected values follow increasing indices. Formats 0–4 use the inherited Zstd wrapper; format 6 uses a direct Zstd frame. This final lossless layer changes the byte representation, not the reconstructed correction values. Format 5 wraps a non-format-5 inner payload. Dimensions and correction counts are stored in record framing.

Let b be the decoded base, P the selected positions, and h a stored channel scale. The Float32 reconstruction operation is $D_h(b, q) =$ `float(b + float(q) * h)`. For quantized formats 4 and 6, decoding obeys

$$\widehat{y}_i = \begin{cases} b_i, & i \notin P, \\ D_h(b_i, q_i), & i \in P,\ q_i \neq 0, \\ v_i, & i \in P,\ q_i = 0, \end{cases} \tag{1}$$

where v_i is a stored exact exception. The q_i are signed integer codes, stored as Int8 in format 4 and bit-packed in format 6; zero at a selected position is an escape. The packed representation maps $q \geq 0$ to $2q$ and $q < 0$ to $-2q - 1$, then uses the bit width needed by the largest mapped code. Format 0 adds Float64 residuals to the promoted base and casts to Float32; format 1 uses Float32 addition followed by exact overrides. Formats 2 and 3 respectively assign replacement values and XOR stored words with base words. Format 5 first fills constant channels, then decodes its nested repair. The decoder needs only base bytes and correction metadata.

A1.1 Search Tie-Breaking and Reuse

Each admissible call returns a repaired candidate and its actual base-plus-repair size. An inadmissible proposal has infinite search cost and still consumes one proposal from the budget. Further refinement selects the adjacent interval with the smallest minimum endpoint cost, breaking ties by larger logarithmic width and then the lower interval. Its geometric midpoint is the next proposal. Stop after at most 22 logical proposals or when no distinct representable midpoint remains. Serf-XOR/QT use divisor 64, five initial proposals, and a limit of eight each. The exact candidate is additional. Equal base-plus-repair sizes favor the larger parameter; remaining ties retain the earlier evaluated candidate. Equivalent effective ZFP precision settings can reuse a physical call, but the actual base error is rechecked against each proposed cap.

Table A1: Correction formats. Fields appear in stored order before any applicable lossless encoding; each payload carries a format tag.

ID	Stored fields
0	Dense Float64 residuals.
1	Dense Float32 residuals; exception mask; exact Float32 values.
2	Position mask; exact Float32 replacements.
3	Position mask; XOR words relative to base Float32 words.
4	Float32 channel scales; position mask; Int8 codes; exact escapes.
5	Override and inner counts; channel mask; Float32 channel constants; nested repair.
6	Code bit width; Float32 channel scales; position mask; packed ZigZag codes; exact escapes.

A1.2 Numerical Boundary Cases

Ordinary Pearson uses denominator floor $\eta = 10^{-15}$, returning zero at or below it. For source–reconstruction Pearson, source centered energy at most η instead gives score one exactly when all decoded Float32 values numerically equal their sources, and zero otherwise. ACF uses the denominator rule. The parameter estimator sets the anchor to zero for low source energy or a zero active ACF gradient, invoking the exact fallback. For $\rho = 1$, repair assigns zero Pearson penalty to an estimate at least one and infinity otherwise. For a degenerate source, it uses $2\mathbf{1}_{m>1}$, with m current source mismatches, while direct acceptance requires $m = 0$.

A2 COMPLETE CONSTRAINT AND CORRECTION ABLATIONS

This record documents the completed October 17, 2026 ablation. Its run identifier and source CSVs are preserved with the artifact. It uses the same 172 inputs, six backend configurations, preprocessing, models, seeds, thresholds, and original simple correction as the native-versus-framework studies of October 15–16. The 27,900 compression configurations contain five arms for each backend,

with 930 configurations per backend/arm. All 5,580 existing ERA configurations and their 43,596 consumer records are reused; the 2,482 Raw consumer records are also reused. Additional input-identical consumer outputs are reused only after identity checks. The complete consumer ledger contains 220,462 records: 206,752 forecasting, 1,488 classification, 1,302 retrieval, and 10,920 anomaly records. These include VAL, Raw, seeds, and verified reuse, not 220,462 independent observations or new inference calls. All expected records are present, the closeout audit passes, and every enabled constraint passes its numerical audit. This study adds no timing panel.

What each comparison changes. E, ER, EA, and ERA retain the same framework search. Omitting R or A disables its contribution to the anchor, repair, and acceptance rule. The E arm is therefore different from calling the native backend once at the user's error bound. E→ER and EA→ERA estimate the effect of enabling the R policy in two contexts; E→EA and ER→ERA do the same for A. These comparisons do not isolate individual formula terms, nor do they hold the adaptive candidate trajectories fixed. All arms additionally undergo a full ERA audit, including their omitted constraints, with the same allowance of $64\epsilon_{\text{mach}}$ for Float64 machine epsilon. Passing an enabled contract and passing this additional full audit are different quantities.

ERA-NC denotes ERA without correction. It replays the stored ordered logical-cap/phase trajectory of the corresponding existing ERA window. The native-cap gate, numerical ERA acceptance, exact Zstd3 candidate, and byte/cap tie rules remain unchanged. Previously unpersisted native candidates are materialized as needed; existing repaired winners are not treated as native candidates. ERA-NC does not adaptively search a new trajectory. The resulting comparison isolates availability of the original simple correction within that candidate pool, conditional on the pool having been generated by the ERA policy. It does not establish the best possible performance of an independently designed algorithm without correction.

Aggregation and units. The full-ablation tables use TEST and retain every prescribed threshold. CR counts the complete JZF1 file, including normalizer, true lengths, channel information, backend and repair payloads, and outer framing. Its numerator is the unpadded Float32 size; NASA's true length is already the TEST suffix length. CR is aggregated geometrically within source and then across sources and settings. Task quality first averages seeds, then inputs within source, sources, settings, and models equally. Solar's 137 series therefore do not receive 137 times the source weight. Counts and pass/fallback rates instead use actual window conditions, including repeated thresholds; they are not source-balanced means. Paired improvement/tie/decline counts describe correlated input–model–threshold conditions, not independent statistical samples.

The forecast metric is MSE (lower is better); classification accuracy, retrieval AP, and anomaly AP/VUS-PR are better when higher. In the constraint-contrast tables, positive changes always mean improvement: MSE is before minus after, and other scores are after minus before, all in their original metric units. Correction CR changes are relative percentages of on versus off, not percentages of stored bytes saved. Repaired-value counts refer to scalar values; repair payload bytes exclude outer framing and are not the denominator of complete CR.

Interpretation and negative results. ER still leaves ACF violations, and EA still leaves original–reconstruction Pearson violations. Thus neither statistic substitutes for the other in the measured joint contract. ERA improves E's macro averages for all four task categories on every backend, but is not universally better than ER or EA. In forecasting, A added to ER worsens mean MSE for SPERR-1D, Serf-XOR, Serf-QT, and SZ2016; R added to EA worsens it for ZFP. Complete constraint protection is the consistent result, not optimal downstream scores or a monotone storage ordering.

Correction has its largest storage benefit for SZ3. Across all 3,024 TEST storage pairs it improves complete CR in 489, ties in 2,295, and reduces it in 240. The original selector compares window-level base-plus-repair bytes, whereas complete JZF1 size also reflects outer framing and encoding; the observed decreases must therefore remain in the result. Forecast MSE is slightly higher with correction on all six backends, with some differences extremely small and no significance claim. No-correction outputs can benefit from more exact fallback or a finer accepted native base. Fallback alone cannot explain every quality difference: both arms have zero forecasting fallback for SPERR-1D, Serf-XOR, Serf-QT, and SZ2016.

These data informed method development. The completed coverage is exploratory evidence, not a preregistered evaluation on unseen data. The two Serf modes are not independent compressor families. The SZ2016 compatibility case retains the documented zlib capacity patch and the historical ASan best-fit-constructor issue; it is not described as a fully sanitizer-clean implementation. No formula/no-formula, density, matched-CR, or new timing study was added in this run.

A3 RETAINED NATIVE-COMPARISON CONSUMER DETAIL

Table A2 retains the original model-level forecasting evidence while the main text gives space to the complete constraint and correction controls. Each entry uses the same frozen model and source-balanced aggregation as the main native comparison. This table is not the E-arm comparison: native calls a backend once, whereas E retains framework search. Small adverse changes for ZFP's Chronos-2 and Chronos-Bolt remain visible in the full-precision source data even where displayed rounding obscures a difference.

A4 RETAINED FORMULA AND SEARCH-DENSITY CONTROL

This earlier 144-condition VAL panel separates data-dependent anchor selection from the overall framework effect. It uses the first series of each forecasting source, its first four validation windows, strict/loose settings, SZ3/ZFP, formula/no-formula initialization, and budgets of 22/44/88 proposals. The no-formula arm substitutes the fixed relative-range geometric midpoint for the data-dependent anchor. It retains candidate refinement, original simple correction, and ERA acceptance; only initial parameter generation ignores statistical sensitivities. Larger budgets extend each policy's own trajectory with the same initial 12 proposals and preserved prefixes.

Table A2: Original forecasting MSE by consumer, averaging sources and settings equally. Each pair is native/JointZip; lower is better. Rounding can hide small differences. Raw uses the same frozen weights. SZ2016 is the subsequent legacy extension.

Consumer	Raw	SZ3	ZFP	SPERR-1D	Serf-XOR	Serf-QT	SZ2016
Ridge	0.129	0.218/0.148	0.135/0.134	0.144/0.139	0.190/0.143	0.195/0.137	0.167/0.139
DLinear	0.097	0.145/0.105	0.100/0.099	0.110/0.105	0.127/0.101	0.126/0.100	0.124/0.101
PatchTST	0.089	0.113/0.092	0.089/0.089	0.097/0.093	0.109/0.092	0.110/0.091	0.110/0.092
Chronos-2	0.116	0.244/0.158	0.126/0.126	0.134/0.130	0.161/0.136	0.160/0.131	0.152/0.130
Chronos-Bolt	0.153	0.255/0.180	0.165/0.166	0.173/0.167	0.190/0.172	0.201/0.175	0.189/0.170
TimesFM 3	0.131	2.133/0.374	0.151/0.147	0.193/0.162	0.186/0.167	0.196/0.194	0.196/0.169

Table A3: Formula versus no-formula search at equal proposal limits. CR changes are paired geometric percentages; MSE changes are ratios of source-balanced means. Higher CR and lower MSE are preferable. Each row has 12 VAL conditions; this is separate from the complete TEST ablation.

Backend	Limit	CR% ↑	Ridge% ↓	DLinear% ↓
SZ3	22	+0.22	+6.05	-0.39
SZ3	44	+0.91	+4.97	-2.41
SZ3	88	+0.30	+10.88	-0.35
ZFP	22	-0.35	+0.00	+0.00
ZFP	44	-0.41	+0.00	+0.00
ZFP	88	-0.39	+0.00	+0.00

For formula-guided SZ3, increasing 22 proposals to 44 improves geometric CR by 0.9576%; 88 yields 0.9577%, largely saturating at 44. ZFP reconstructions and scores remain unchanged, with CR changes below 0.04%. Additional proposals need not monotonically shrink complete files because window selection, parameter records, and outer compression affect their byte ordering.

At equal limits, formula-guided SZ3 has 0.22–0.91% higher CR than the no-formula control, whereas ZFP has 0.35–0.41% lower CR. SZ3's Ridge MSE is worse with the formula anchor, while DLinear is slightly better. Formula guidance has a small, inconsistent incremental advantage here, not an explanation of all native-versus-framework gains. These frozen-weight CPU VAL diagnostics do not replace the main GPU TEST scores. Cached incremental times are excluded from speed comparisons.

Table A4: Forecasting complete-stream CR (larger is better), using source-balanced geometric means over all prescribed TEST thresholds. E retains framework search. ERA-NC reuses the ordered ERA candidate trajectory.

Backend	E	ER	EA	ERA	ERA-NC
SZ3	4.7604	3.1126	3.6882	3.0814	2.7322
ZFP	7.1224	5.1646	5.6117	5.0842	5.0820
SPERR-1D	16.9601	10.0206	11.3117	9.6851	9.6914
Serf-XOR	14.2132	6.0772	6.9591	5.9393	5.8907
Serf-QT	23.6507	11.0781	11.6043	10.6534	10.6496
SZ2016	10.7395	5.7647	6.7683	5.6274	5.5565

Table A5: Forecasting full numerical ERA pass rate (%). These rates audit all three constraints even when some were disabled. All arms pass their enabled constraints; the audit uses the fixed $64\epsilon_{\text{mach}}$ tolerance. Rates are ratios of window counts, not source-macro averages.

Backend	E	ER	EA	ERA	ERA-NC
SZ3	52.97	88.54	71.99	100.00	100.00
ZFP	82.10	90.26	95.77	100.00	100.00
SPERR-1D	52.03	76.13	82.64	100.00	100.00
Serf-XOR	19.41	63.24	72.49	100.00	100.00
Serf-QT	20.14	65.75	75.65	100.00	100.00
SZ2016	33.61	69.35	72.26	100.00	100.00

Table A6: UCR complete-stream CR (larger is better), using source-balanced geometric means over all prescribed TEST thresholds. E retains framework search. ERA-NC reuses the ordered ERA candidate trajectory.

Backend	E	ER	EA	ERA	ERA-NC
SZ3	18.5899	7.7062	12.2791	7.6938	6.5647
ZFP	12.7623	11.8832	12.5576	11.7702	11.7071
SPERR-1D	31.6427	23.6014	28.1726	23.4873	23.4873
Serf-XOR	29.9735	12.9754	18.8859	12.8747	12.5970
Serf-QT	47.5833	27.8006	26.5844	27.6784	27.4759
SZ2016	17.7657	10.1359	13.2043	10.0782	9.9517

Table A7: UCR full numerical ERA pass rate (%). These rates audit all three constraints even when some were disabled. All arms pass their enabled constraints; the audit uses the fixed $64\epsilon_{\text{mach}}$ tolerance. Rates are ratios of window counts, not source-macro averages.

Backend	E	ER	EA	ERA	ERA-NC
SZ3	1.99	98.13	5.10	100.00	100.00
ZFP	66.67	98.01	69.28	100.00	100.00
SPERR-1D	32.59	97.01	37.06	100.00	100.00
Serf-XOR	2.99	97.89	9.70	100.00	100.00
Serf-QT	0.87	99.25	11.94	100.00	100.00
SZ2016	6.97	97.26	12.31	100.00	100.00

Table A8: NASA complete-stream CR (larger is better), using source-balanced geometric means over all prescribed TEST thresholds. E retains framework search. ERA-NC reuses the ordered ERA candidate trajectory.

Backend	E	ER	EA	ERA	ERA-NC
SZ3	7.5198	7.0325	7.2769	6.9459	6.8213
ZFP	9.0260	8.6012	8.6160	8.5321	8.5321
SPERR-1D	14.7540	13.2834	13.7605	12.9934	12.9911
Serf-XOR	14.3680	11.1404	11.4894	10.9192	10.9571
Serf-QT	22.0295	17.1211	17.5712	16.4503	16.4412
SZ2016	11.8077	10.2094	10.3901	10.0080	9.9938

Table A9: NASA full numerical ERA pass rate (%). These rates audit all three constraints even when some were disabled. All arms pass their enabled constraints; the audit uses the fixed $64\epsilon_{\text{mach}}$ tolerance. Rates are ratios of window counts, not source-macro averages.

Backend	E	ER	EA	ERA	ERA-NC
SZ3	82.21	94.21	88.35	100.00	100.00
ZFP	87.42	99.14	96.21	100.00	100.00
SPERR-1D	57.27	87.42	76.85	100.00	100.00
Serf-XOR	49.62	91.00	76.31	100.00	100.00
Serf-QT	41.19	84.78	67.38	100.00	100.00
SZ2016	58.34	90.57	78.81	100.00	100.00

Table A10: Forecast MSE. Lower is better. Seeds are averaged first; sources and models then receive equal weight. Values pool all prescribed TEST thresholds. ERA-NC uses the fixed ERA trajectory, so this is not an independently optimized no-correction algorithm.

Backend	Raw	E	ER	EA	ERA	ERA-NC
SZ3	0.119221	0.541603	0.181931	0.239893	0.176107	0.175690
ZFP	0.119221	0.127167	0.127125	0.126488	0.126934	0.126934
SPERR-1D	0.119221	0.140371	0.131600	0.133310	0.132747	0.132724
Serf-XOR	0.119221	0.156495	0.134716	0.138410	0.135152	0.134092
Serf-QT	0.119221	0.162950	0.134706	0.142802	0.138150	0.138149
SZ2016	0.119221	0.155321	0.133126	0.139564	0.133301	0.133299

Table A11: Classification accuracy. Higher is better. Seeds are averaged first; sources and models then receive equal weight. Values pool all prescribed TEST thresholds. ERA-NC uses the fixed ERA trajectory, so this is not an independently optimized no-correction algorithm.

Backend	Raw	E	ER	EA	ERA	ERA-NC
SZ3	0.738730	0.593581	0.657941	0.597500	0.657385	0.666800
ZFP	0.738730	0.686953	0.689290	0.687045	0.689800	0.689654
SPERR-1D	0.738730	0.629477	0.641062	0.629919	0.640910	0.640910
Serf-XOR	0.738730	0.576697	0.644772	0.601395	0.648753	0.650385
Serf-QT	0.738730	0.578938	0.659450	0.591303	0.659417	0.660316
SZ2016	0.738730	0.557316	0.640005	0.564563	0.639681	0.643276

Table A12: Retrieval mAP. Higher is better. Seeds are averaged first; sources and models then receive equal weight. Values pool all prescribed TEST thresholds. ERA-NC uses the fixed ERA trajectory, so this is not an independently optimized no-correction algorithm.

Backend	Raw	E	ER	EA	ERA	ERA-NC
SZ3	0.755221	0.640746	0.690642	0.655377	0.690416	0.696071
ZFP	0.755221	0.711505	0.713781	0.711769	0.714064	0.713948
SPERR-1D	0.755221	0.681627	0.687159	0.682249	0.687591	0.687591
Serf-XOR	0.755221	0.626238	0.687421	0.649387	0.689208	0.691662
Serf-QT	0.755221	0.627572	0.696986	0.638319	0.697334	0.697050
SZ2016	0.755221	0.611324	0.679332	0.629294	0.679418	0.681323

Table A13: NASA AP. Higher is better. Seeds are averaged first; sources and models then receive equal weight. Values pool all prescribed TEST thresholds. ERA-NC uses the fixed ERA trajectory, so this is not an independently optimized no-correction algorithm.

Backend	Raw	E	ER	EA	ERA	ERA-NC
SZ3	0.233097	0.226856	0.230408	0.232470	0.231316	0.231988
ZFP	0.233097	0.231707	0.234361	0.233938	0.234267	0.234267
SPERR-1D	0.233097	0.231671	0.234574	0.234304	0.235146	0.235154
Serf-XOR	0.233097	0.224486	0.238516	0.234125	0.238428	0.238430
Serf-QT	0.233097	0.241210	0.252452	0.251845	0.252458	0.252474
SZ2016	0.233097	0.227397	0.235529	0.232923	0.234225	0.234225

Table A14: NASA VUS-PR. Higher is better. Seeds are averaged first; sources and models then receive equal weight. Values pool all prescribed TEST thresholds. ERA-NC uses the fixed ERA trajectory, so this is not an independently optimized no-correction algorithm.

Backend	Raw	E	ER	EA	ERA	ERA-NC
SZ3	0.278552	0.271056	0.274771	0.276223	0.275421	0.276181
ZFP	0.278552	0.276772	0.279457	0.278983	0.279427	0.279427
SPERR-1D	0.278552	0.276348	0.279659	0.279495	0.280271	0.280279
Serf-XOR	0.278552	0.267821	0.282377	0.279746	0.282900	0.282891
Serf-QT	0.278552	0.287307	0.301985	0.300547	0.302141	0.302158
SZ2016	0.278552	0.270793	0.278967	0.277855	0.278893	0.278907

Table A15: Forecast MSE incremental constraint effects. Top entries are MSE reductions or score increases in raw units (positive is better); bottom entries count improved/tied/declined input–model–threshold pairs after seed averaging. These correlated pairs are not independent samples. Pair counts and source-macro effects have different weighting.

Backend	$E \rightarrow ER$	$E \rightarrow EA$	$ER \rightarrow ERA$	$EA \rightarrow ERA$
	+0.359672	+0.301710	+0.005824	+0.063786
SZ3	1778/0/778	1571/0/985	1385/246/925	1656/0/900
	+0.000042	+0.000679	+0.000191	-0.000447
ZFP	1117/503/936	1332/115/1109	1200/302/1054	681/1431/444
	+0.008771	+0.007062	-0.001147	+0.000563
SPERR-1D	1327/0/1229	1367/0/1189	1316/0/1240	1265/0/1291
	+0.021779	+0.018084	-0.000436	+0.003258
Serf-XOR	1740/0/816	1711/0/845	1458/0/1098	1454/0/1102
	+0.028244	+0.020147	-0.003444	+0.004652
Serf-QT	1753/0/803	1721/0/835	1411/0/1145	1473/0/1083
	+0.022194	+0.015756	-0.000175	+0.006263
SZ2016	1717/0/839	1648/0/908	1393/7/1156	1571/1/984

Table A16: Classification accuracy incremental constraint effects. Top entries are MSE reductions or score increases in raw units (positive is better); bottom entries count improved/tied/declined input–model–threshold pairs after seed averaging. These correlated pairs are not independent samples. Pair counts and source-macro effects have different weighting.

Backend	$E \rightarrow ER$	$E \rightarrow EA$	$ER \rightarrow ERA$	$EA \rightarrow ERA$
	+0.064360	+0.003919	-0.000556	+0.059885
SZ3	14/2/8	12/4/8	0/22/2	14/3/7
	+0.002338	+0.000093	+0.000509	+0.002754
ZFP	5/18/1	3/19/2	3/20/1	5/18/1
	+0.011585	+0.000442	-0.000152	+0.010991
SPERR-1D	13/7/4	6/11/7	3/19/2	11/8/5
	+0.068075	+0.024698	+0.003981	+0.047358
Serf-XOR	16/2/6	13/6/5	5/18/1	14/4/6
	+0.080512	+0.012365	-0.000033	+0.068114
Serf-QT	16/2/6	14/4/6	1/21/2	18/2/4
	+0.082689	+0.007247	-0.000324	+0.075118
SZ2016	14/2/8	12/5/7	1/21/2	17/3/4

Table A17: Retrieval mAP incremental constraint effects. Top entries are MSE reductions or score increases in raw units (positive is better); bottom entries count improved/tied/declined input–model–threshold pairs after seed averaging. These correlated pairs are not independent samples. Pair counts and source-macro effects have different weighting.

Backend	$E \rightarrow ER$	$E \rightarrow EA$	$ER \rightarrow ERA$	$EA \rightarrow ERA$
	+0.049895	+0.014631	-0.000225	+0.035039
SZ3	14/0/4	12/0/6	3/9/6	13/0/5
	+0.002276	+0.000264	+0.000283	+0.002295
ZFP	7/9/2	5/9/4	6/9/3	7/9/2
	+0.005531	+0.000622	+0.000432	+0.005342
SPERR-1D	12/3/3	8/3/7	7/6/5	10/3/5
	+0.061183	+0.023149	+0.001787	+0.039821
Serf-XOR	13/0/5	15/0/3	5/12/1	12/0/6
	+0.069414	+0.010746	+0.000349	+0.059016
Serf-QT	14/0/4	12/0/6	7/9/2	14/0/4
	+0.068008	+0.017970	+0.000086	+0.050124
SZ2016	15/0/3	11/0/7	4/13/1	16/0/2

Table A18: NASA AP incremental constraint effects. Top entries are MSE reductions or score increases in raw units (positive is better); bottom entries count improved/tied/declined input–model–threshold pairs after seed averaging. These correlated pairs are not independent samples. Pair counts and source-macro effects have different weighting.

Backend	$E \rightarrow ER$	$E \rightarrow EA$	$ER \rightarrow ERA$	$EA \rightarrow ERA$
	+0.003552	+0.005614	+0.000908	-0.001154
SZ3	57/124/35	55/129/32	44/150/22	53/125/38
	+0.002655	+0.002232	-0.000094	+0.000329
ZFP	29/170/17	38/161/17	22/182/12	11/198/7
	+0.002903	+0.002633	+0.000572	+0.000842
SPERR-1D	89/65/62	95/64/57	70/93/53	82/68/66
	+0.014030	+0.009639	-0.000088	+0.004303
Serf-XOR	93/68/55	94/66/56	59/99/58	92/77/47
	+0.011242	+0.010635	+0.000006	+0.000613
Serf-QT	84/66/66	104/50/62	90/77/49	79/76/61
	+0.008132	+0.005526	-0.001304	+0.001302
SZ2016	79/82/55	77/70/69	56/101/59	69/95/52

Table A19: NASA VUS-PR incremental constraint effects. Top entries are MSE reductions or score increases in raw units (positive is better); bottom entries count improved/tied/declined input–model–threshold pairs after seed averaging. These correlated pairs are not independent samples. Pair counts and source-macro effects have different weighting.

Backend	$E \rightarrow ER$	$E \rightarrow EA$	$ER \rightarrow ERA$	$EA \rightarrow ERA$
	+0.003715	+0.005167	+0.000650	-0.000802
SZ3	56/124/36	52/129/35	39/150/27	56/124/36
	+0.002686	+0.002211	-0.000031	+0.000444
ZFP	35/169/12	41/162/13	18/185/13	12/198/6
	+0.003310	+0.003146	+0.000613	+0.000776
SPERR-1D	87/62/67	97/63/56	67/93/56	78/66/72
	+0.014556	+0.011925	+0.000523	+0.003154
Serf-XOR	98/66/52	100/66/50	71/99/46	91/76/49
	+0.014678	+0.013239	+0.000156	+0.001594
Serf-QT	88/66/62	101/51/64	90/79/47	83/76/57
	+0.008174	+0.007062	-0.000074	+0.001038
SZ2016	72/81/63	71/70/75	53/102/61	71/95/50

Table A20: Forecasting TEST mechanism counts. Fallback and repaired percentages use windows as denominator. Repaired values are scalar values. Base and repair bytes are selected payload counters, not complete JZF1 size; the CR tables include framing and metadata.

Backend	Arm	Windows	Exact (%)	Repaired (%)	Values	Base bytes	Repair bytes
SZ3	E	13632	27.11	0.00	0	2420874	0
SZ3	ER	13632	44.54	7.53	39365	2523884	51419
SZ3	EA	13632	38.03	8.86	9329	2474837	39499
SZ3	ERA	13632	46.79	8.06	40259	2532436	54026
SZ3	ERA-NC	13632	50.97	0.00	0	2606221	0
ZFP	E	13632	0.51	0.00	0	1518318	0
ZFP	ER	13632	0.87	0.01	98	1538984	50
ZFP	EA	13632	1.47	0.49	503	1545774	2155
ZFP	ERA	13632	1.50	0.12	171	1554477	481
ZFP	ERA-NC	13632	1.52	0.00	0	1554991	0
SPERR-1D	E	13632	0.00	0.00	0	562272	0
SPERR-1D	ER	13632	0.00	0.00	0	615759	0
SPERR-1D	EA	13632	0.00	1.09	851	616081	4540
SPERR-1D	ERA	13632	0.00	0.22	287	634979	946
SPERR-1D	ERA-NC	13632	0.00	0.00	0	636092	0
Serf-XOR	E	13632	0.00	0.00	0	791348	0
Serf-XOR	ER	13632	0.00	0.90	5435	959076	6140
Serf-XOR	EA	13632	0.00	5.49	11653	948515	26906
Serf-XOR	ERA	13632	0.00	1.67	10617	999610	10290
Serf-XOR	ERA-NC	13632	0.00	0.00	0	1011820	0
Serf-QT	E	13632	0.00	0.00	0	363368	0
Serf-QT	ER	13632	0.00	0.00	0	411174	0
Serf-QT	EA	13632	0.00	0.15	218	431269	754
Serf-QT	ERA	13632	0.00	0.02	83	439316	130
Serf-QT	ERA-NC	13632	0.00	0.00	0	439474	0
SZ2016	E	13632	0.00	0.00	0	775375	0
SZ2016	ER	13632	0.00	0.72	6807	878035	5462
SZ2016	EA	13632	0.00	4.91	9334	869247	22771
SZ2016	ERA	13632	0.00	2.55	12839	905628	14957
SZ2016	ERA-NC	13632	0.00	0.00	0	923737	0

Table A21: UCR TEST mechanism counts. Fallback and repaired percentages use windows as denominator. Repaired values are scalar values. Base and repair bytes are selected payload counters, not complete JZF1 size; the CR tables include framing and metadata.

Backend	Arm	Windows	Exact (%)	Repaired (%)	Values	Base bytes	Repair bytes
SZ3	E	804	0.62	0.00	0	154757	0
SZ3	ER	804	1.24	37.81	36216	178432	35153
SZ3	EA	804	0.62	11.19	474	160282	2968
SZ3	ERA	804	1.24	37.94	36245	178478	35195
SZ3	ERA-NC	804	3.98	0.00	0	240594	0
ZFP	E	804	3.23	0.00	0	180945	0
ZFP	ER	804	3.23	2.99	203	192473	1122
ZFP	EA	804	4.10	0.62	7	181179	141
ZFP	ERA	804	4.10	3.61	210	192635	1263
ZFP	ERA-NC	804	4.48	0.00	0	194246	0
SPERR-1D	E	804	0.00	0.00	0	69996	0
SPERR-1D	ER	804	0.00	0.00	0	98181	0
SPERR-1D	EA	804	0.00	4.60	106	72019	1178
SPERR-1D	ERA	804	0.00	0.00	0	98276	0
SPERR-1D	ERA-NC	804	0.00	0.00	0	98276	0
Serf-XOR	E	804	2.61	0.00	0	206818	0
Serf-XOR	ER	804	3.11	28.11	50445	223366	40168
Serf-XOR	EA	804	2.61	21.64	1103	213233	6374
Serf-XOR	ERA	804	3.11	28.36	50559	223339	40284
Serf-XOR	ERA-NC	804	3.11	0.00	0	276080	0
Serf-QT	E	804	0.00	0.00	0	59189	0
Serf-QT	ER	804	0.00	7.09	452	88470	2652
Serf-QT	EA	804	0.00	5.35	88	62031	1299
Serf-QT	ERA	804	0.00	7.09	452	88481	2652
Serf-QT	ERA-NC	804	0.00	0.00	0	92693	0
SZ2016	E	804	0.00	0.00	0	106531	0
SZ2016	ER	804	0.00	23.13	41939	124463	35011
SZ2016	EA	804	0.00	25.37	5442	112907	15289
SZ2016	ERA	804	0.00	23.63	42055	124460	35178
SZ2016	ERA-NC	804	0.00	0.00	0	164046	0

Table A22: NASA TEST mechanism counts. Fallback and repaired percentages use windows as denominator. Repaired values are scalar values. Base and repair bytes are selected payload counters, not complete JZF1 size; the CR tables include framing and metadata.

Backend	Arm	Windows	Exact (%)	Repaired (%)	Values	Base bytes	Repair bytes
SZ3	E	2799	69.56	0.00	0	362336	0
SZ3	ER	2799	70.24	4.64	2959	368259	5493
SZ3	EA	2799	70.13	2.93	375	366093	2557
SZ3	ERA	2799	70.60	6.40	3590	368416	7303
SZ3	ERA-NC	2799	71.20	0.00	0	382109	0
ZFP	E	2799	29.55	0.00	0	266835	0
ZFP	ER	2799	40.48	0.00	0	272097	0
ZFP	EA	2799	40.48	0.00	0	271505	0
ZFP	ERA	2799	40.48	0.00	0	272385	0
ZFP	ERA-NC	2799	40.48	0.00	0	272385	0
SPERR-1D	E	2799	14.26	0.00	0	143395	0
SPERR-1D	ER	2799	31.33	0.00	0	151216	0
SPERR-1D	EA	2799	31.76	0.21	18	149321	176
SPERR-1D	ERA	2799	32.05	0.18	42	153306	169
SPERR-1D	ERA-NC	2799	32.05	0.00	0	153505	0
Serf-XOR	E	2799	33.55	0.00	0	186742	0
Serf-XOR	ER	2799	34.48	2.72	3556	207723	3997
Serf-XOR	EA	2799	34.48	2.57	965	202793	2393
Serf-XOR	ERA	2799	34.76	2.97	3560	209284	4193
Serf-XOR	ERA-NC	2799	35.01	0.00	0	214424	0
Serf-QT	E	2799	16.65	0.00	0	94722	0
Serf-QT	ER	2799	16.65	0.00	0	102440	0
Serf-QT	EA	2799	17.47	1.07	89	101889	887
Serf-QT	ERA	2799	17.47	0.32	15	105613	242
Serf-QT	ERA-NC	2799	17.47	0.00	0	105930	0
SZ2016	E	2799	11.83	0.00	0	178784	0
SZ2016	ER	2799	28.65	0.57	622	199143	626
SZ2016	EA	2799	28.83	1.46	1348	194740	1415
SZ2016	ERA	2799	28.87	1.11	819	200723	1187
SZ2016	ERA-NC	2799	28.87	0.00	0	202275	0

Table A23: Correction on versus off on the stored ERA candidate trajectory. Full CR gains are $100(\mathrm{CR}_{\mathrm{on}}/\mathrm{CR}_{\mathrm{off}} - 1)$; all metadata are billed. Counts compare complete TEST files. Total improved/tied/declined: 489/2295/240. Both sides pass full numerical ERA.

Backend	Task	Off CR	On CR	Gain (%)	I/T/D
SZ3	Forecasting	2.7322	3.0814	+12.7802	190/148/88
SZ3	UCR	6.5647	7.6938	+17.1986	5/1/0
SZ3	NASA	6.8213	6.9459	+1.8274	19/52/1
ZFP	Forecasting	5.0820	5.0842	+0.0441	6/412/8
ZFP	UCR	11.7071	11.7702	+0.5385	1/4/1
ZFP	NASA	8.5321	8.5321	0.0000	0/72/0
SPERR-1D	Forecasting	9.6914	9.6851	-0.0650	14/399/13
SPERR-1D	UCR	23.4873	23.4873	0.0000	0/6/0
SPERR-1D	NASA	12.9911	12.9934	+0.0176	3/68/1
Serf-XOR	Forecasting	5.8907	5.9393	+0.8261	73/292/61
Serf-XOR	UCR	12.5970	12.8747	+2.2052	4/2/0
Serf-XOR	NASA	10.9571	10.9192	-0.3463	5/61/6
Serf-QT	Forecasting	10.6496	10.6534	+0.0360	2/423/1
Serf-QT	UCR	27.4759	27.6784	+0.7368	1/5/0
Serf-QT	NASA	16.4412	16.4503	+0.0553	4/67/1
SZ2016	Forecasting	5.5565	5.6274	+1.2761	152/215/59
SZ2016	UCR	9.9517	10.0782	+1.2714	3/3/0
SZ2016	NASA	9.9938	10.0080	+0.1425	7/65/0

Table A24: Correction on-minus-off task differences in raw metric units. Positive MSE means deterioration; positive accuracy, AP, and VUS-PR mean improvement. Scientific notation preserves tiny effects. No significance claim follows from these aggregate differences.

Backend	Δ MSE	Δ Acc.	Δ mAP	Δ NASA AP	Δ VUS-PR
SZ3	$+4.166\times10^{-4}$	-9.415×10^{-3}	-5.654×10^{-3}	-6.720×10^{-4}	-7.596×10^{-4}
ZFP	$+2.896\times10^{-7}$	$+1.454\times10^{-4}$	$+1.168\times10^{-4}$	0	0
SPERR-1D	$+2.308\times10^{-5}$	0	0	-7.967×10^{-6}	-8.283×10^{-6}
Serf-XOR	$+1.060\times10^{-3}$	-1.632×10^{-3}	-2.454×10^{-3}	-1.653×10^{-6}	$+8.918\times10^{-6}$
Serf-QT	$+1.060\times10^{-6}$	-8.993×10^{-4}	$+2.847\times10^{-4}$	-1.611×10^{-5}	-1.671×10^{-5}
SZ2016	$+2.029\times10^{-6}$	-3.595×10^{-3}	-1.906×10^{-3}	$+1.141\times10^{-7}$	-1.379×10^{-5}